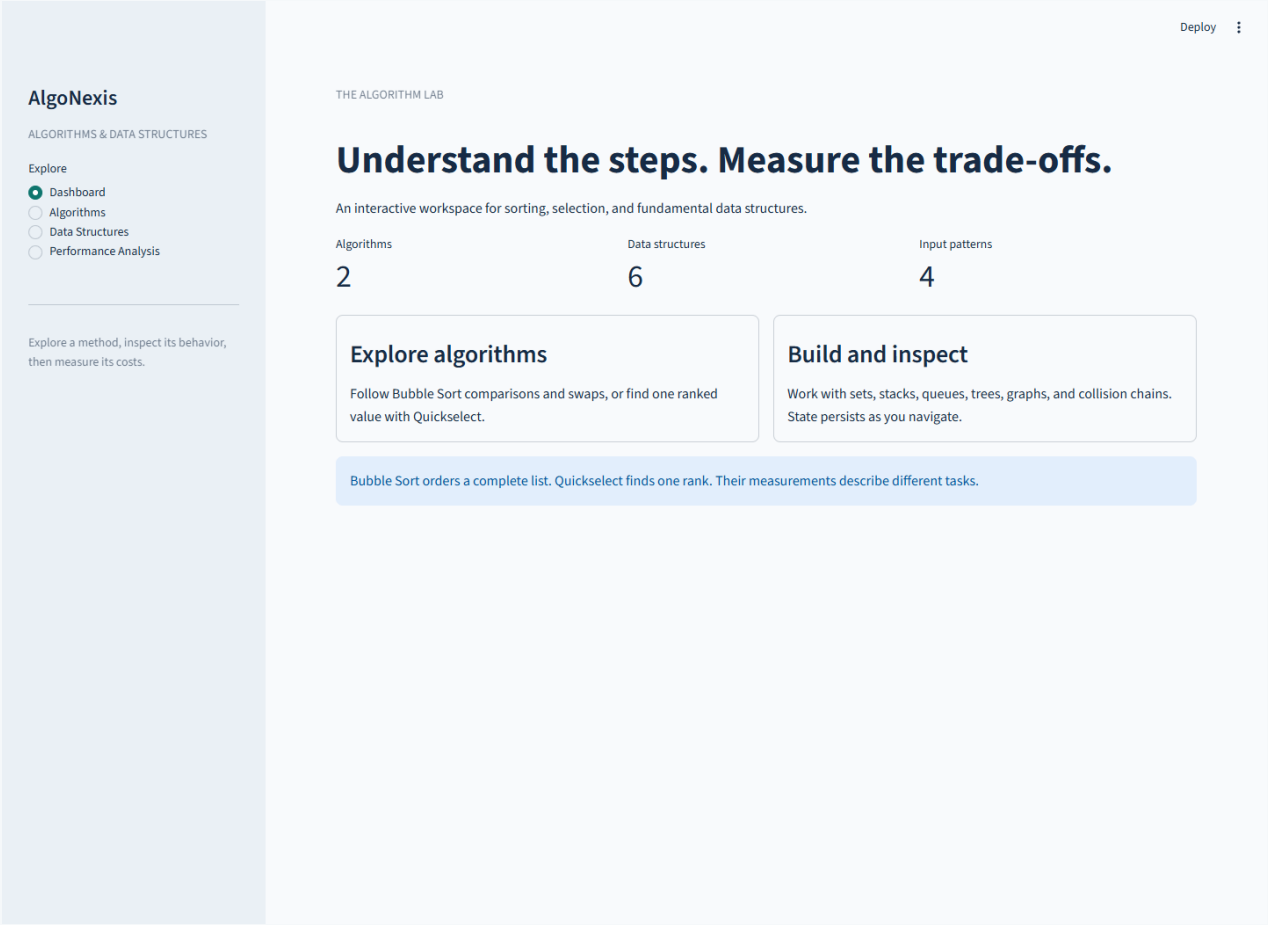
AlgoNexis

An interactive algorithms & data structures portfolio

Shreyash · CSU Global · CSC506

November 7, 2026 · Portfolio v1.0.0



Source: AlgoNexis implementation and recorded measurements

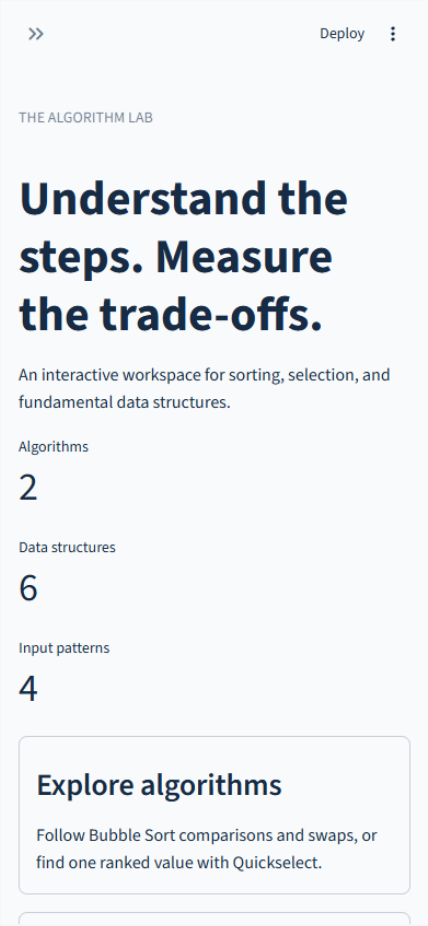
One portfolio, four ways to explore

Dashboard: scope and guide

Algorithms: sort, select, inspect steps

Data Structures: six persistent models

Performance: measurements and CSV



Source: AlgoNexis implementation and recorded measurements

Architecture keeps the core independently testable

Streamlit views + session state

Bounded input · explicit errors · independent model state

Algorithms → immutable frames

Bubble Sort · Quickselect · step retrieval

Manager → six structures

Explicit registry · persistent instances

Performance → CSV → figures → report / slides

Fresh fixtures · correctness checks · separate time and allocation trials

Source: AlgoNexis implementation and recorded measurements

Bubble Sort: make every comparison inspectable

Adjacent compare and swap

Early exit: O(n) best case

O(n²) average / worst, untraced

Immutable snapshots; trace limit 50



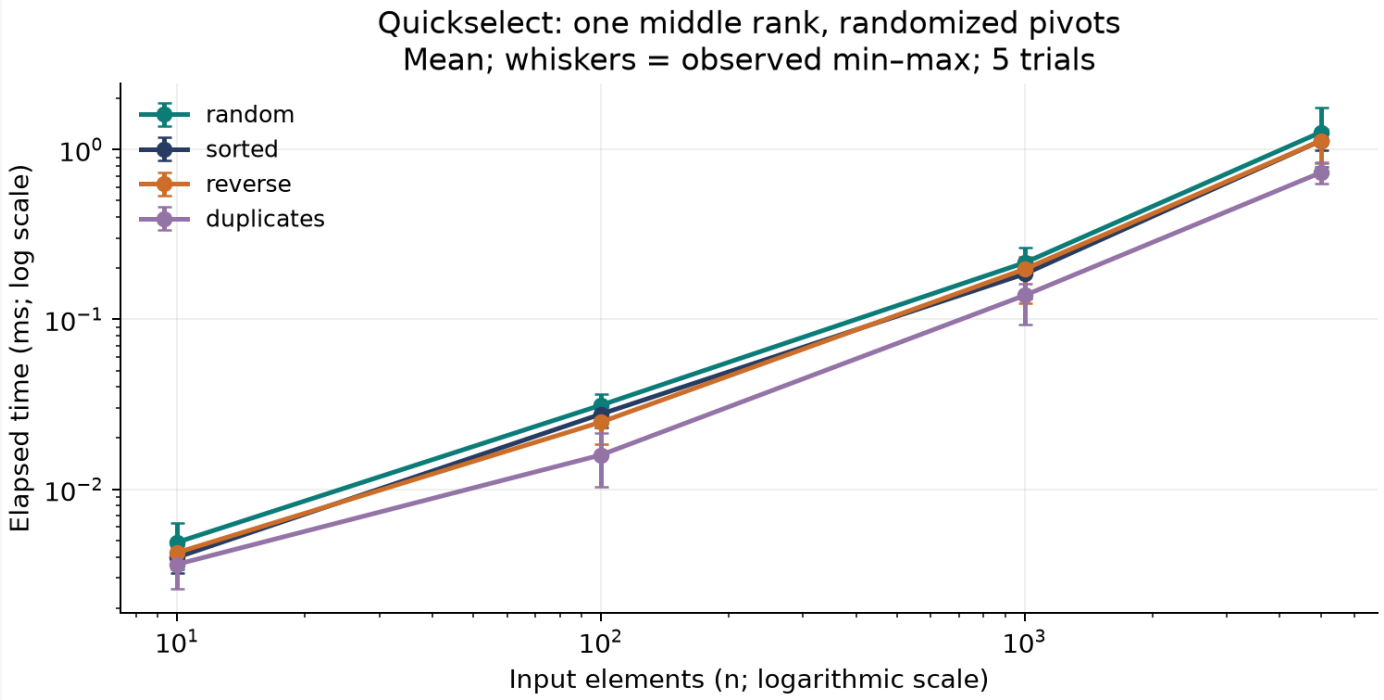
Quickselect: ask for one rank

One-based k; duplicates retain ranks

Randomized three-way partition

Expected O(n); worst O(n²)

Copy preserves the caller's input



Custom Set: membership defines equality

- Dictionary-backed membership
- Explicit union / intersection loops
- Difference and symmetric difference
- Derived results preserve both operands

Stack and Queue: order determines representation

Stack: list end → LIFO, amortized O(1) push

Queue: deque ends → FIFO, O(1) removal

Empty removal raises a clear IndexError

Draining n elements is an O(n) batch

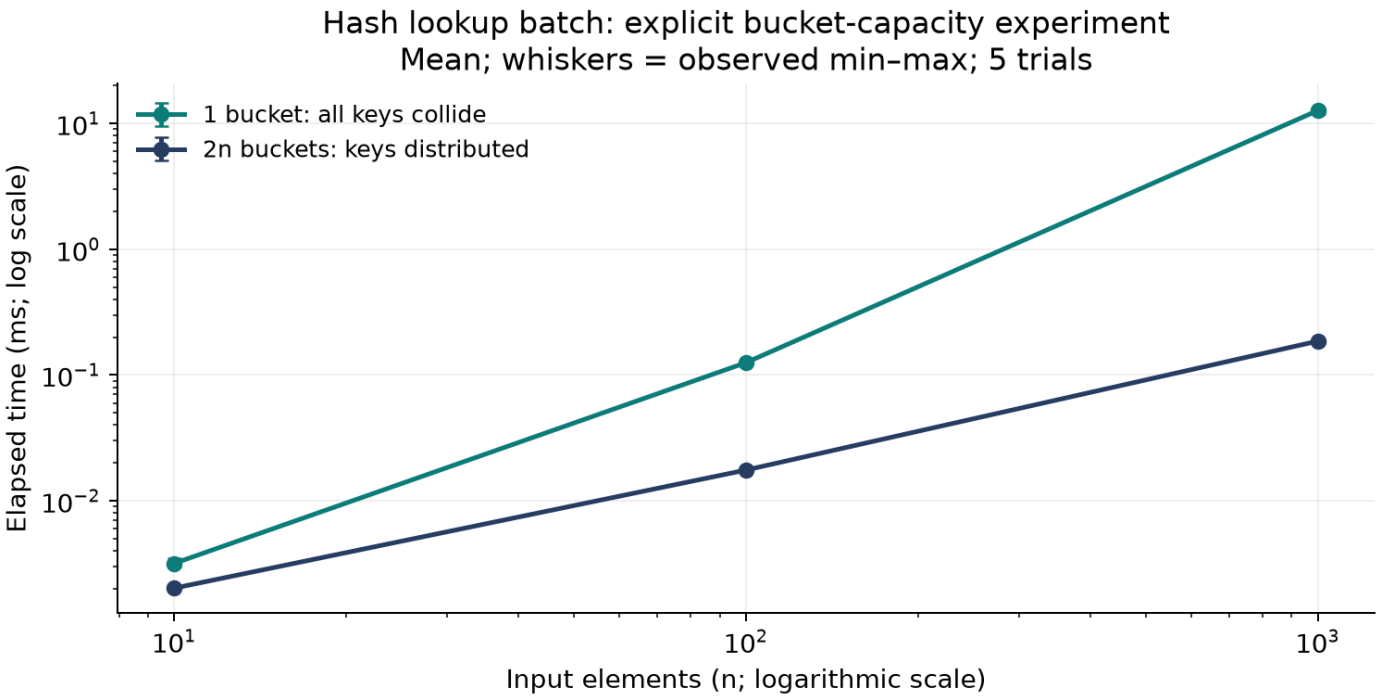
Tree, Graph, Hash Table: expose the trade-offs

BST: ordered keys; shape controls search

Graph: undirected adjacency; BFS / DFS

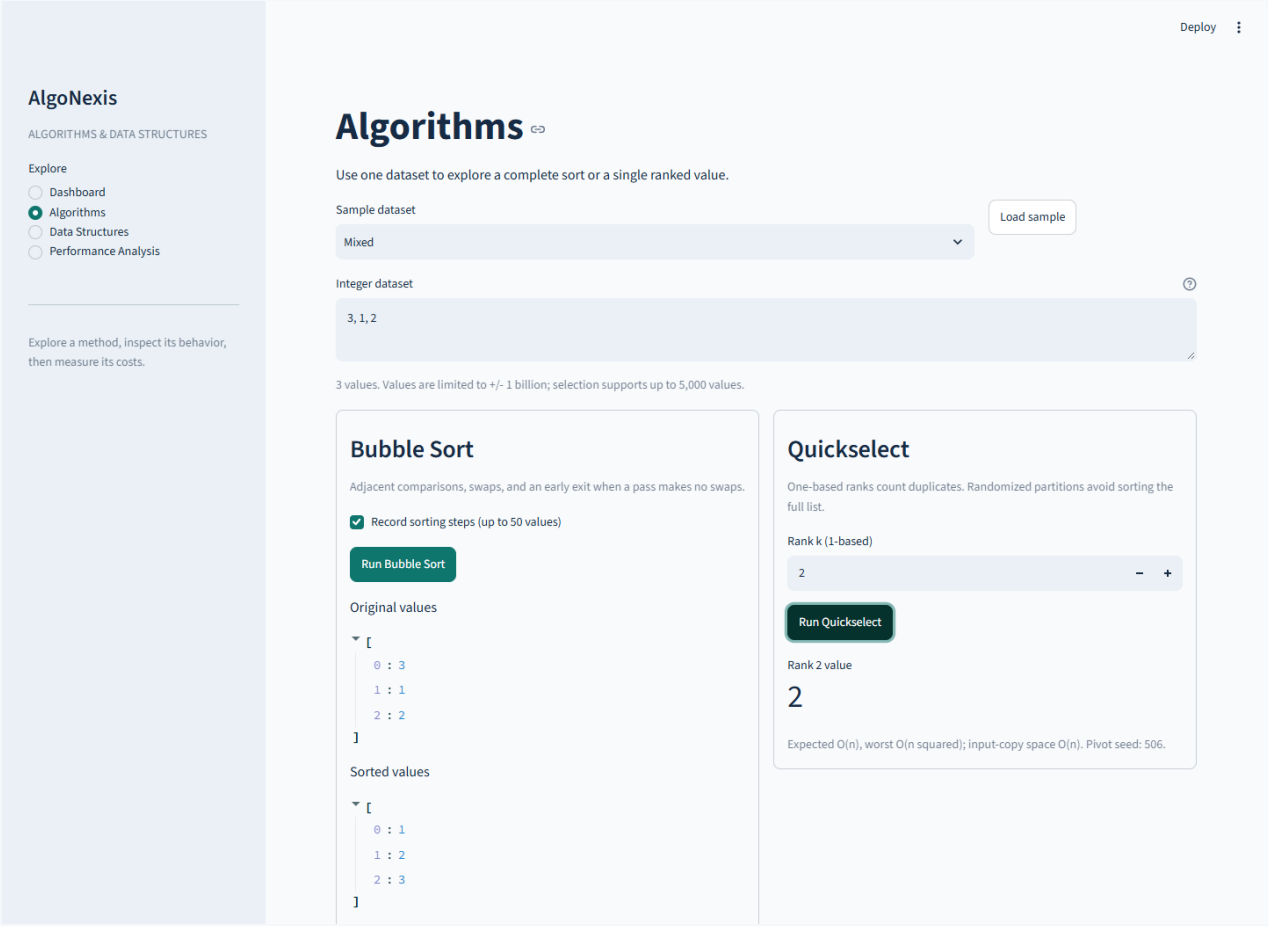
Hash Table: fixed separate chains

One bucket / 2n lookup means: 68.20×



The interface is part of correctness

- Strict integer parsing; no eval
- Separate model and widget state
- Explicit growth and trace limits
- Navigation, failures, resets checked in Edge



Source: AlgoNexis implementation and recorded measurements

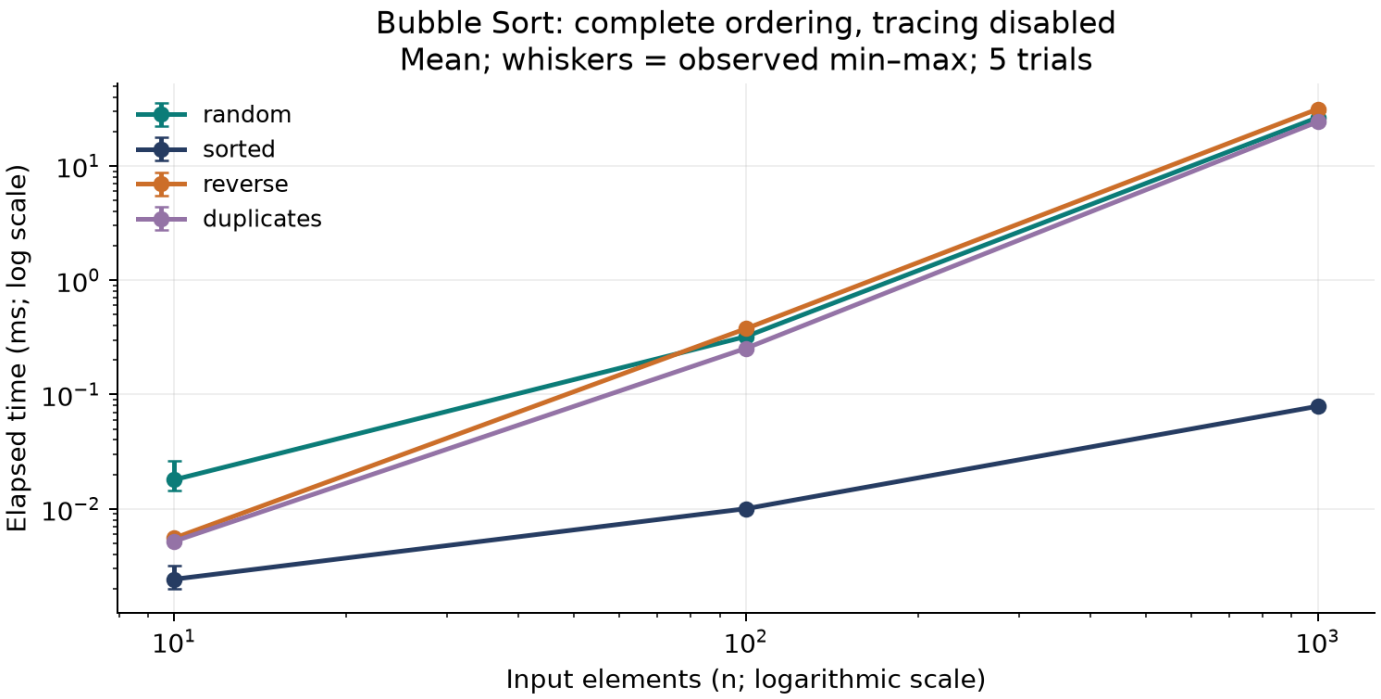
Measure the work that the label describes

1,080 observed rows + 8 guarded skips

Five timing + five allocation repeats per case

Setup / checking excluded; output included

Separate 60-row controlled collision profile



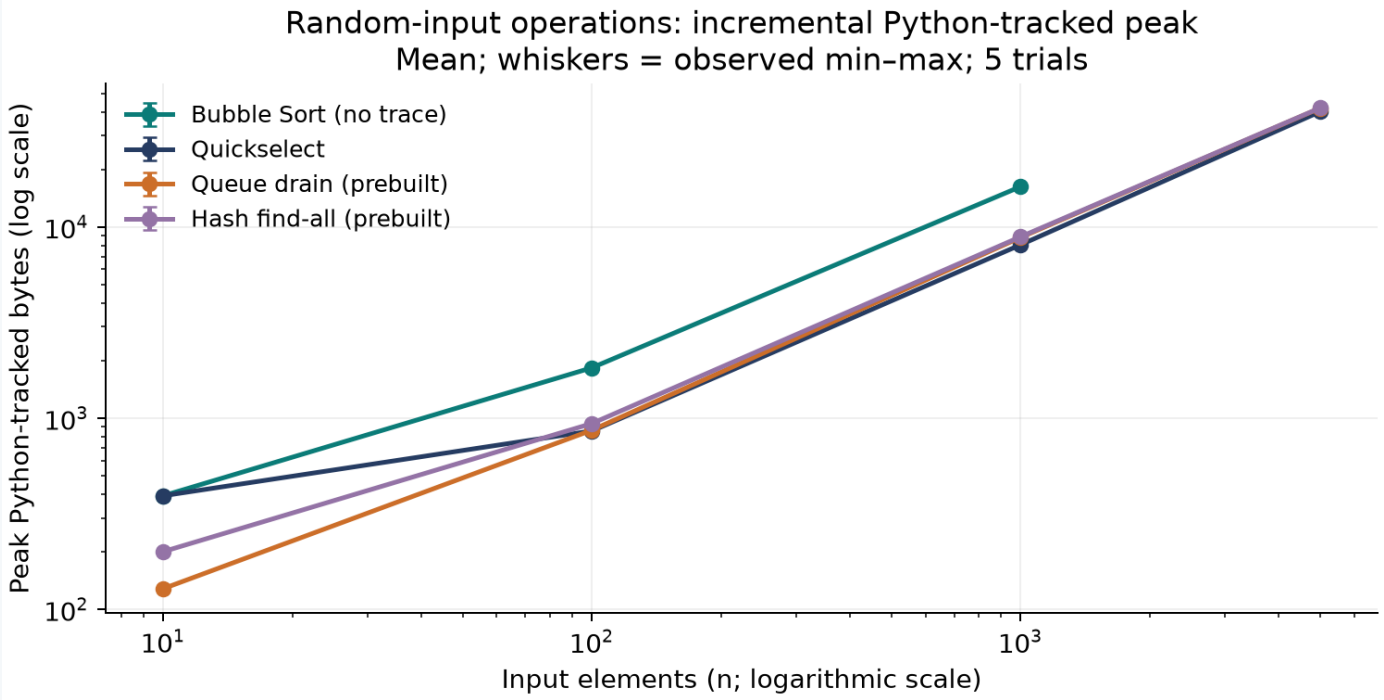
Findings need boundaries

BST sorted / random means: 26.47×

Repeated keys reduce effective fixture size

Tracked operation allocation ≠ total RAM

One machine, five repeats, selected workloads



Python Software Foundation, tracemalloc; final measured summaries; prebuilt fixtures excluded

Learning outcomes and release

Contracts connect algorithms, UI, tests, and evidence

195 automated tests + real browser walkthrough

Report: 8 substantive pages; slides: 12

Next: balanced tree, resizing hash, broader datasets

Source: AlgoNexis implementation and recorded measurements